

Welcome to Code Crunch!

C10 | CRUNCH LABS: Crunching BioInformatics

Fri 3-4PM Weekly | Spring 2025

Attendance

Earn a Certificate of
Completion by attending
at least 70% of the
workshops



linktr.ee/CODE.CRUNCH





Unit #0

C10 | CRUNCH LABS: Crunching BioInformatics

Unit 0: Introducing Coding Relevance to Biology

What is BioInformatics



What is BioInformatics



Bioinformatics is the study of Biology using coding methods

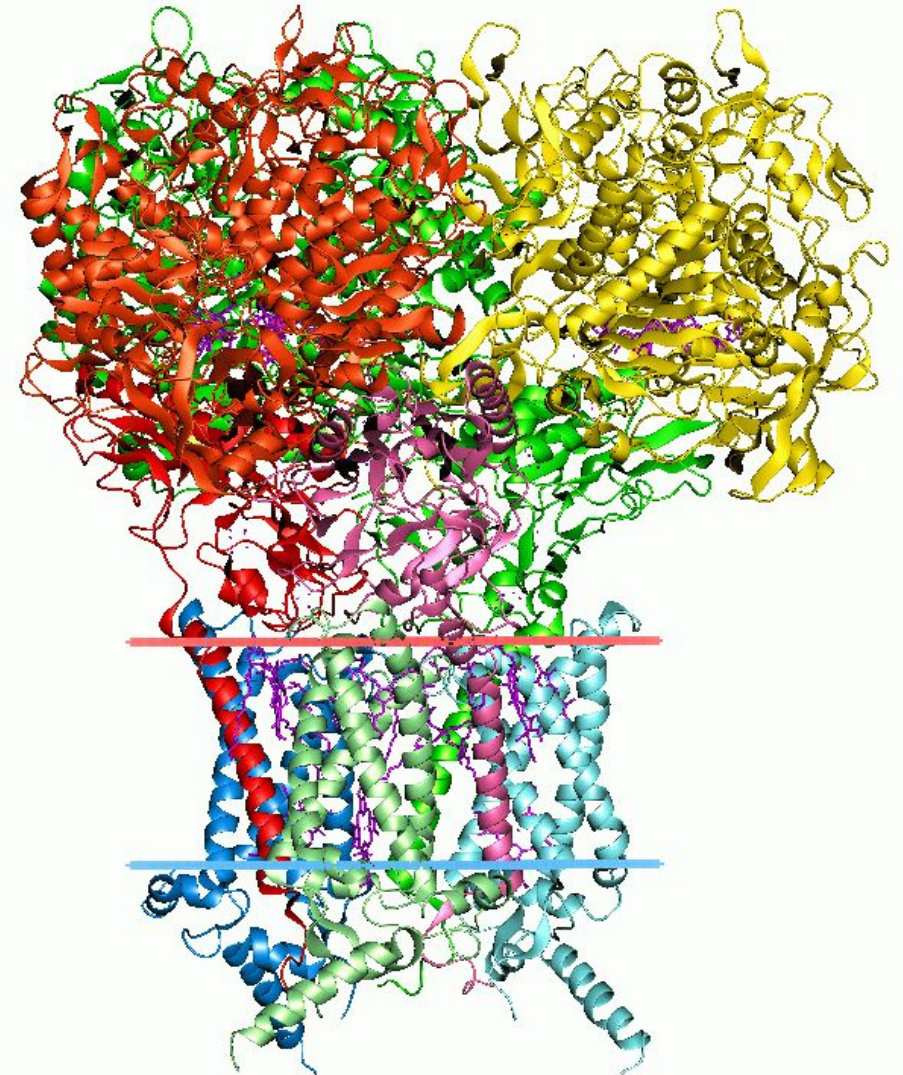
Python, Complex Biological Systems

DNA, Proteins, Polymers

Icebreaker/Discussion



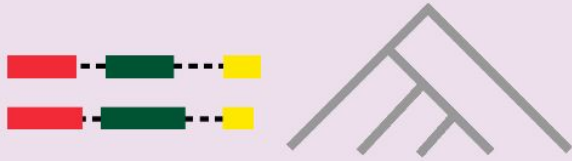
What ways are
Biology and
Coding
Interconnected?



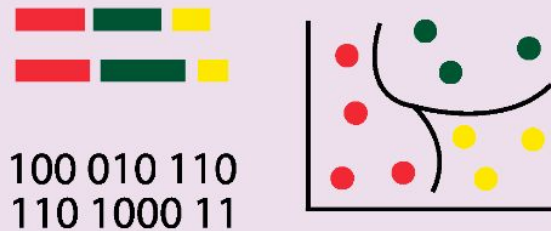
Machine and deep learning integration with bioinformatics

Molecular evolution

Phylogenetic inference



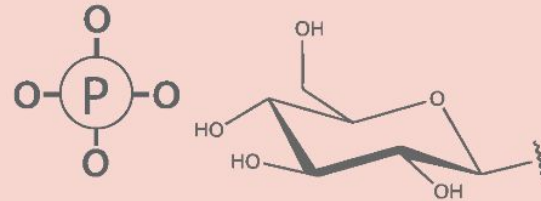
*Alignment-free
sequence classification*



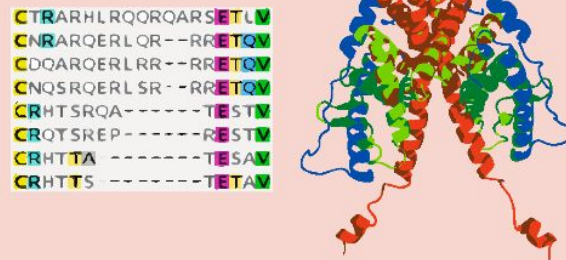
- Inference of tree topology
- Sequence classification
- Viral sequence identification
- functional annotation

Protein structure Analysis

Post translational modification



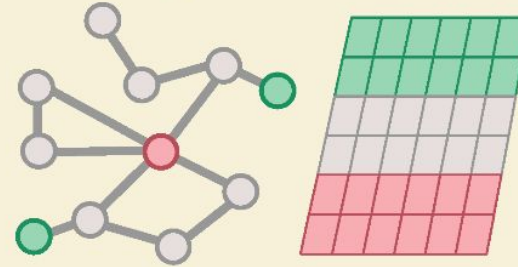
Folding and structure



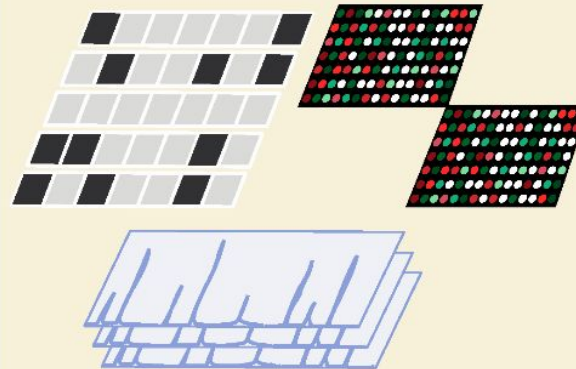
- Phosphorylation site prediction
- Protein glycosylation prediction
- Protein contact maps prediction
- Structural homology prediction

Systems biology

Biological Networks



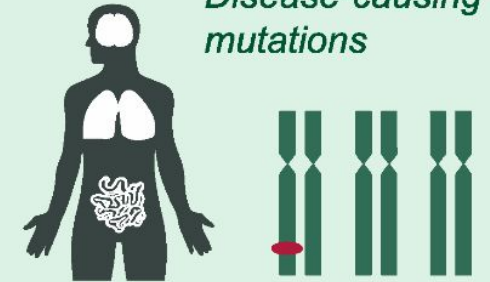
Multi-Omics integration



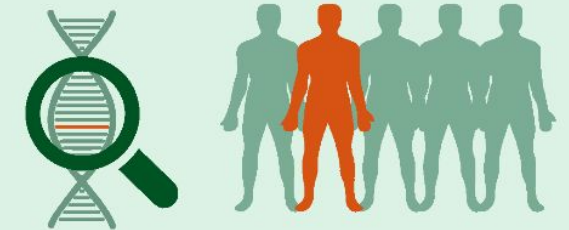
- Biological networks construction
- Biological interactions prediction
- Pathway dynamics prediction
- Platform integration frameworks

Genomics for Disease Research

Disease-causing mutations



Biomarkers discovery



- Disease associated genes and mutations
- Biomarkers
- Precision medicine applications

BioInformatics Revelance



Studying Disease and
Cancer

Algorithms to improve
treatment

For Genome-Wide
Association Studies



CRUNCHING BioInformatics

Agenda:

Unit 0: Introducing Coding Relevance to Biology

Unit 1: Introduction of Bioinformatics

Unit 2: Python for Bioinformatics

Unit 3: Sequence Analysis

Unit 4: Genomics and Proteomics

Unit 5: Modeling Biological Systems

Unit 6: Machine Learning in Bioinformatics

Unit 7: Ethical Considerations in Biomedical Data

Unit 8: Mini-Project Development

What is our goal?



DNA Sequencing

Modeling Biological Systems

Using Functions to facilitate
information gathering

Installations

- Python
- Jupyter Notebook
- Biopython



Python



- 1) Python.org
- 2) Downloads
- 3) Different Versions displayed (latest version will pop up, click on it)
- 4) After download, you can open the exe. file and click on the box with “add python.exe to PATH” and then click “Install Now”
- 5) After setup is successful, You can open it and even interact with the command prompt



Jupyter Notebook

1. Type terminal in search bar, and command prompt should open.
2. Type “python --version” (to check the version and if you have downloaded python successfully
3. Type download jupyter notebook on the browser.
4. Type Terminal in search bar, in the command prompt type: “pip install notebook”
5. After installation, to run Jupyter notebook, in the terminal/windows powershell type either: “jupyter notebook”
or “python -m notebook” (if previous didn’t work)
6. Tab should open up with ability to go into browser and see the notebook. Add a new file called notebook and
in select Kernel, click Python 3. Select it and type a print() to test
7. Save the notebook in by clicking on File -> Save Notebook.
8. To rename file click the file name which is next to the Jupyter logo.

Biopython



- 1) In the terminal type: “pip install biopython”
- 2) After it downloads successfully, its ready!



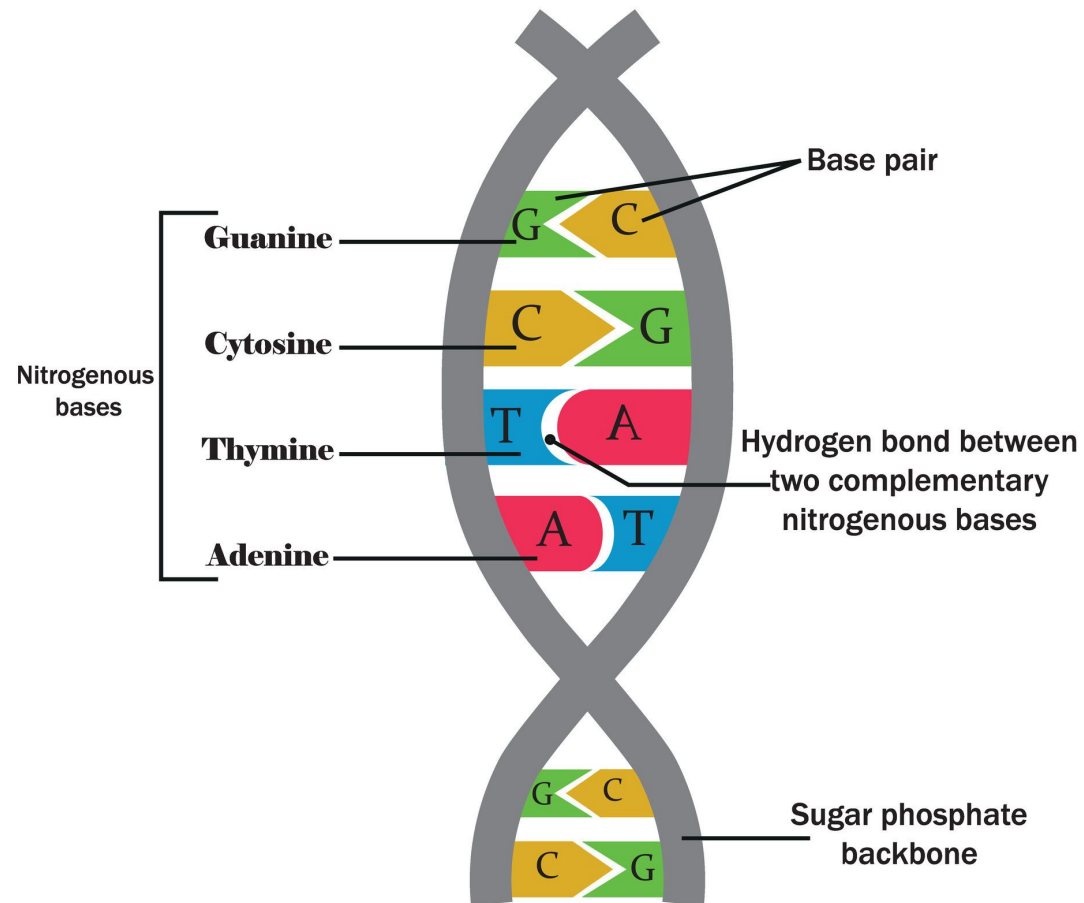
Google Collab (Backup)

1. Type Google Colab
2. Sign in
3. Create new notebook
4. Start typing code!

DNA Nucleotides

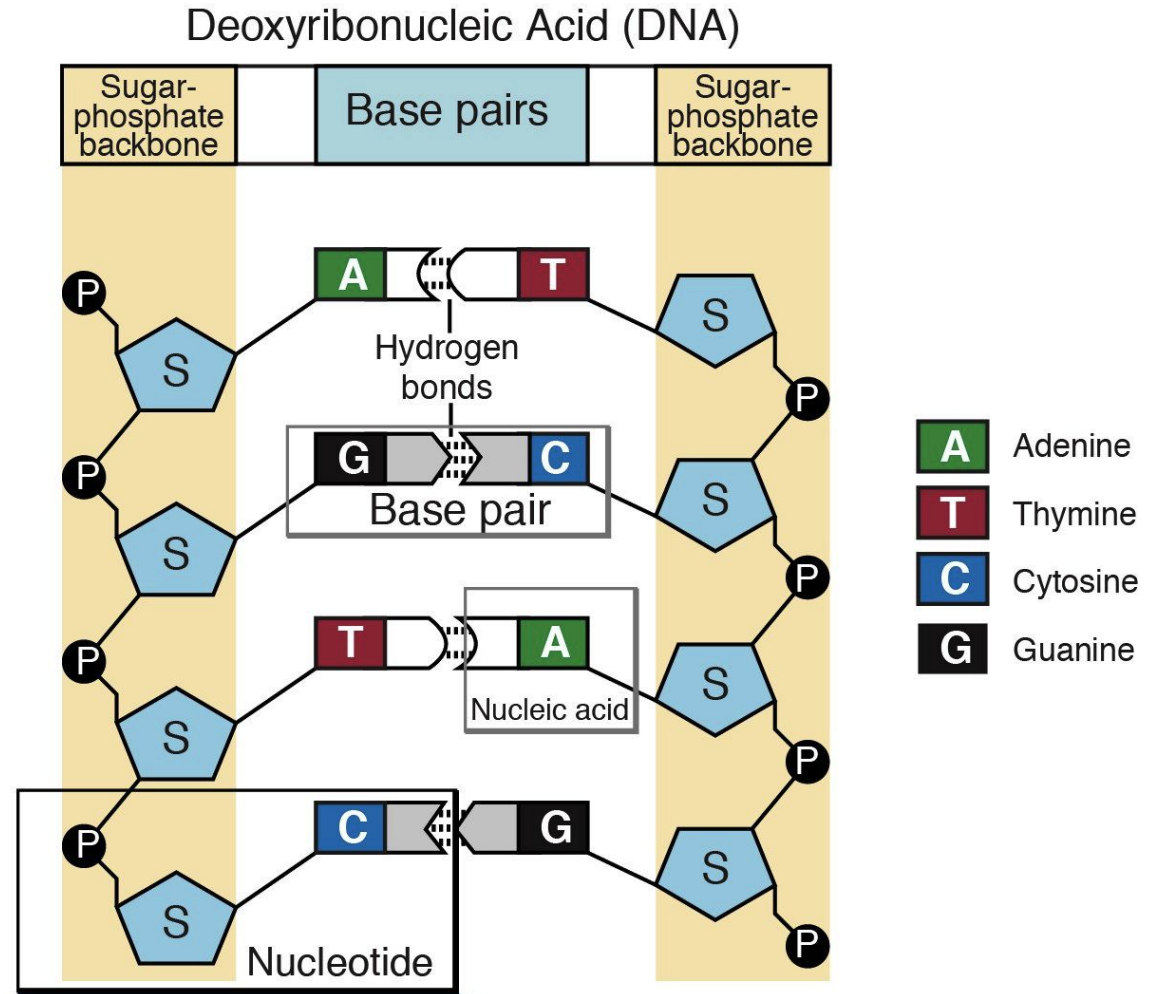
1. Adenine
2. Guanine
3. Cytosine
4. Thymine

DNA STRUCTURE



DNA Nucleotides

- Ribose, Sugar, Nucleotides
- Phosphate Groups
- 4 Base Pairs, in RNA one of them is different.





Q&A Session



Attendance



Join us for the upcoming **weekly** sessions this Spring 2025!



Monday: C5 1PM / C3 2PM
Tuesdays: C7 1PM
Wednesdays: C5 1PM / C1 2PM
Thursdays: C6 1PM / C4 2PM / C8 3PM
Friday: C2 1PM / C9 2PM / C9 3PM



RSVP LU.MA/CODECRUNCH



Your feedback helps us improve. Share your thoughts!

Go to our website: go.fiu.edu/codecrunch